Foundation model for EEG with residual vector quantized autoencoder

Yanic ROTHLINGSHOFER

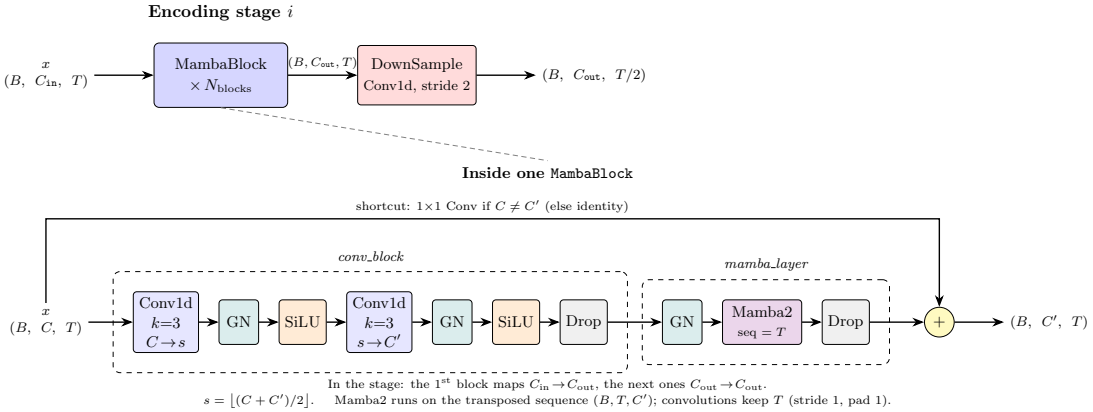
Télécom Paris

Supervisor : Van Tam Nguyen

Outline

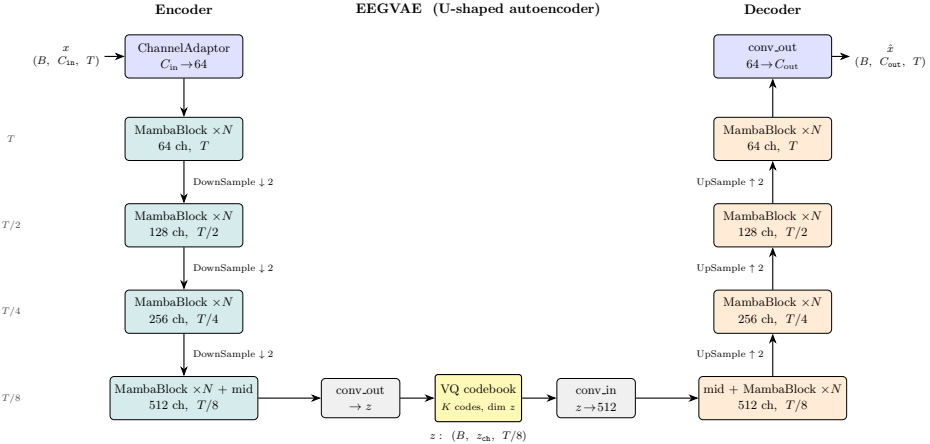
1. EEG-RQVAE architecture
2. Pretraining objectives
3. Downstream evaluation

Encoding stage — Mamba block



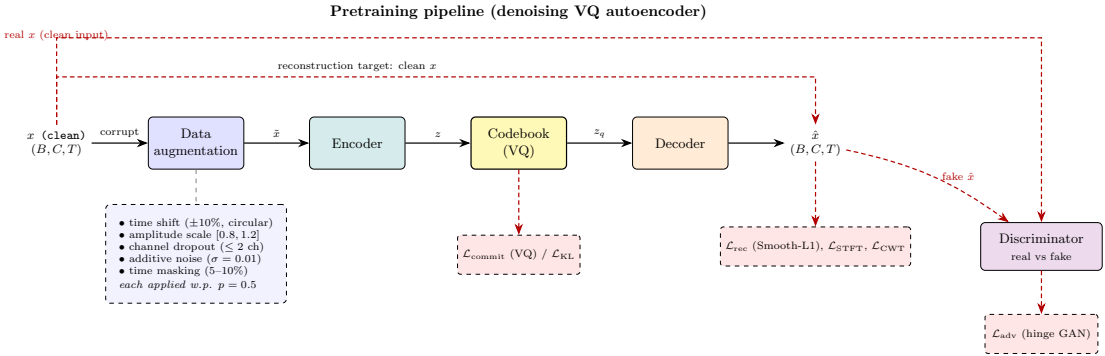
N_{blocks} residual conv + Mamba2 blocks per stage, then a stride-2 downsampling.

Model pipeline — U-shaped autoencoder



Bottleneck = quantize z to the nearest codebook entries (VQ) *or* a Gaussian posterior (KL).
Decoder mirrors the encoder — **no skip connections** (information passes only through z).

Pretraining pipeline — denoising VQ autoencoder



$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{stft}}\mathcal{L}_{\text{STFT}} + \lambda_{\text{cwt}}\mathcal{L}_{\text{CWT}} + \lambda_{\text{lat}}\left(\mathcal{L}_{\text{commit}}/\mathcal{L}_{\text{KL}}\right) + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}}$$

Multi-resolution STFT loss

Compares reconstruction $\hat{x}$ and target x in the time–frequency domain, over several FFT resolutions $\mathcal{S} = \{64, 128, 256, 512\}$ (Hann window, hop $= n/4$):

$$\mathcal{L}_{\text{STFT}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \underbrace{\frac{\big\| \, |S_s \hat{x}| - |S_s x| \, \big\|_F}{\big\| \, |S_s x| \, \big\|_F}}_{\text{spectral convergence}} + \underbrace{\big\| \log |S_s \hat{x}| - \log |S_s x| \big\|_1}_{\text{log-magnitude}}$$

where $S_s x$ is the STFT of x at resolution s (magnitude only).

Advantages

- ✓ Captures spectral content (EEG rhythms $\delta, \theta, \alpha, \beta, \gamma$) missed by time-domain L1.
- ✓ Multi-resolution: both time and frequency detail.
- ✓ Cheap and differentiable (FFT).

Drawbacks

- ✗ Magnitude only: phase is ignored.
- ✗ Fixed time/frequency trade-off per window.
- ✗ Sensitive to window/hop choices.

Continuous Wavelet Transform (CWT) loss

Same idea as STFT but with complex Morlet wavelets ψ_a at scales $a \in A$ tuned to EEG bands (δ to γ); $W_a x = x * \psi_a$:

$$\mathcal{L}_{\text{CWT}} = \frac{1}{|A|} \sum_{a \in A} \frac{\left\| \, |W_a \hat{x}| - |W_a x| \, \right\|_F}{\left\| \, |W_a x| \, \right\|_F} + \left\| \log |W_a \hat{x}| - \log |W_a x| \right\|_1$$

Advantages

- ✓ Adaptive time–frequency resolution: good for non-stationary EEG and transients.
- ✓ Scales aligned with physiological bands.
- ✓ Better low-frequency localisation than fixed STFT.

Drawbacks

- ✗ Computationally heavier (convolution at many scales).
- ✗ Sensitive to mother wavelet / scale choice.
- ✗ Currently disabled ($\lambda_{\text{cwt}} = 0$) for cost.

Adversarial loss (hinge GAN)

A discriminator D learns to tell real EEG x from reconstructions $\hat{x}$; the autoencoder (generator G) tries to fool it:

$$\mathcal{L}_D = \tfrac{1}{2}\,\mathbb{E}_x\big[\max(0,\,1 - D(x))\big] + \tfrac{1}{2}\,\mathbb{E}_{\hat{x}}\big[\max(0,\,1 + D(\hat{x}))\big]$$

$$\mathcal{L}_{\text{adv}}^{G} = -\,\mathbb{E}_{\hat{x}}\big[D(\hat{x})\big]$$

Advantages

- ✓ Sharper, more realistic reconstructions (avoids the over-smoothing of L1/MSE).
- ✓ Learns a data-driven perceptual criterion.

Drawbacks

- ✗ Training instability: needs a small weight / warm-up.
- ✗ Risk of mode collapse.
- ✗ Extra discriminator to train and tune.

Downstream results (BACC)

Model	BCI2A	PHYSIONETMI	SLEEPEDFX	SEEDV	FACED	TUAB
EEG-VAE (VQ-Mamba)	–	–	–	–	–	–
EEG-VAE (KL-Mamba)	–	–	–	–	–	–
EEG-VAE (VQ-Transformer)	–	–	–	–	–	–
CBraMod	–	–	–	–	–	–
EEGMamba	–	–	–	–	–	–
EEGPT	–	–	–	–	–	–
LUNA	–	–	–	–	–	–
FEMBA	–	–	–	–	–	–

Balanced accuracy (BACC), linear-probing / channel-training — to be filled.